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Research Paper

A machine learning framework for residential district cooling: Forecasting consumption, explaining drivers, and evaluating decarbonization pathways

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ABSTRACT

District Cooling Systems (DCS) in the Middle East, while energy-efficient, are significant contributors to carbon emissions. This study introduces a novel framework to decarbonize DCS operations by integrating predictive machine learning, explainable AI (XAI), and renewable energy planning, all grounded in extensive real-world data. Leveraging a unique dataset from 59 residential buildings in the UAE—including energy consumption, climate variables, and building features—we developed a high-fidelity cooling load forecasting model. Following a rigorous chronological validation methodology, the Random Forest model was identified as the most robust, achieving a strong performance ($R^2 = 0.8256$, RMSE = 11,668.31). Outdoor temperature was confirmed as a primary driver of cooling load ($r = 0.79$). XAI analysis using SHAP confirmed that temperature, month, and occupancy rate are the most influential predictors. Crucially, scenario-based evaluations found that, under annual-energy assumptions without temporal matching, a 25 % solar offset yields greater carbon savings than an aggressive 30 % energy efficiency improvement. These findings demonstrate a scalable, data-driven blueprint for enhancing the operational efficiency and sustainability of residential DCS. The framework provides an interpretable, evidence-based tool for policymakers to support climate goals, such as the UAE Vision 2050, in arid urban regions.

1. Introduction

In the context of rapid urbanization and population growth in the Middle East, particularly in the United Arab Emirates (UAE), energy demand in residential buildings has been rising. Cooling load consumption—the energy required to maintain indoor thermal comfort—is a major component of this demand, especially in hot and humid climates in the area. High ambient temperatures necessitate extensive use of air conditioning systems, significantly contributing to residential energy consumption. This increased cooling demand leads to higher carbon emissions, a key environmental challenge, as energy production in the region remains largely fossil fuel-dependent (Gupta et al., 2023).

The urgency of reducing carbon emissions has grown with global climate concerns. In the UAE, the residential sector is a major contributor to overall energy consumption and emissions (Eveloy and Ayou, 2019). In the Emirate of Dubai, the residential sector is a major contributor to overall energy consumption. In 2024, residential buildings accounted for 30.62 % of total electricity consumption (DEWA, ANNUAL STATISTICS, 2024), with a significant portion driven by cooling needs. As such, reducing cooling load is pivotal to meeting the UAE Vision 2050 targets for a low-carbon, energy-efficient economy (MOEI, 2023). The strategy emphasizes improving energy efficiency and increasing the share of renewable energy sources in the energy mix, with a particular focus on solar power, given the UAE's favorable climatic

Abbreviations: 5GDHC, fifth-generation district heating and cooling; AI, Artificial Intelligence; BUA, Built-Up Area; CaDI, Carbon Database Initiative; CO_2, Carbon dioxide; DCS, District Cooling Systems; DEWA, Dubai Electricity and Water Authority; ECMWF, European Centre for Medium-Range Weather Forecasts; EDA, exploratory data analysis; EE, energy efficiency; GB, Gradient Boosting; HVAC, Heating, Ventilation, and Air Conditioning; KWh, kilowatt-hours; LR, Linear Regression; MAE, Mean Absolute Error; ML, Machine Learning; MOEI, Ministry of Energy and Infrastructure; MSE, Mean Squared Error; PV, photovoltaic; RF, Random Forest; RH, Relative Humidity; RMSE, Root Mean Squared Error; SHAP, SHapley Additive exPlanations; Ssrd, Surface Solar Radiation Downwards; T2m, 2-meter Air Temperature; TRH, Ton-hours of Refrigeration; UAE, United Arab Emirates; VRF, Variable Refrigerant Flow; XGBoost/XGB, Extreme Gradient Boosting.

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conditions (Haleem et al., 2024).

District cooling is a key energy solution for urban areas with high cooling demands. By centralizing chilled water production and distributing it to multiple buildings via insulated pipes, it offers greater energy efficiency, reduces peak electricity load, and lowers environmental impact. In dense residential zones, it outperforms conventional cooling by cutting energy use while providing reliable, consistent service (Sen et al., 2017). However, inefficient load planning remains a persistent challenge. Overestimation of cooling requirements—often driven by optimistic real estate projections and conservative engineering safety factors—has resulted in oversized cooling infrastructure, leading to excess capacity costs for developers, operators, and consumers (Eveloy and Ayou, 2019).

Enhancing its performance requires a deep understanding of the factors that influence cooling load consumption. Residential buildings in UAE, with their diverse typologies, floor areas, and occupancy patterns, present complex variability in cooling demand (Giusti and Almoosawi, 2017). This research explores how various factors influence cooling load consumption in residential buildings, affecting the energy efficiency and carbon impact of district cooling systems (DCS). Yet, most traditional forecasting approaches either rely on simplified assumptions or complex building simulations, which may not scale effectively for district-level planning. Moreover, many existing studies rely on synthetic or simulated data rather than actual operational data, limiting their practical applicability (Abdel-Jaber and Dirks, 2024).

To address the critical gap left by simulation-based and single-building studies, this research introduces a novel framework grounded in a rare, large-scale empirical dataset. We leverage a unique dataset of actual energy consumption from a cohort of 59 residential buildings serviced by a single district cooling system (DCS) in Dubai, integrated with corresponding environmental and architectural features. This rich, empirical foundation enables the development of a machine learning model that robustly captures the complex, real-world dynamics of urban cooling demand. To ensure our model is not a "black box," we integrate SHapley Additive Explanations (SHAP) to provide transparent insights into the key drivers of energy consumption, fostering stakeholder trust. Furthermore, the framework quantifies the carbon reduction potential by modeling the integration of solar photovoltaics (PV) at 25 %, 50 %, and 75 % penetration levels. By uniquely combining large-scale, district-level empirical data with explainable AI, this work provides a scalable, evidence-based pathway for decarbonizing district cooling systems in line with the UAE's sustainability objectives.

2. Literature review

Climate change poses significant challenges for urban areas, particularly in hot climates where rising temperatures and extreme heat events increase energy demands for cooling. In cities of UAE, cooling demand represents a major share of total electricity consumption, primarily driven by air conditioning systems. Extensive research has explored factors influencing cooling loads, including building typology, floor area, and occupancy patterns (Delzendeh et al., 2017; Kamal and Ahmed, 2023; Santamouris and Vasilakopoulou, 2021). Additionally, predictive modeling techniques using machine learning have been investigated to enhance forecasting accuracy (Bassi et al., 2021; Gao et al., 2024; Ertosun Yıldız and Beyhan, 2025).

District Cooling Systems (DCS), which provide centralized chilled water to multiple buildings, offer a scalable and energy-efficient alternative to traditional cooling, and numerous studies have assessed their performance by integrating efficiency, environmental impact, economic viability, and technological advancements to optimize operations (Bai et al., 2024; Gang et al., 2016). The study (Salimi and Al-Ghamdi, 2020) highlights the importance of integrating DCS into urban infrastructure to mitigate the effects of rising temperatures. By utilizing district cooling, cities can reduce energy consumption, lower greenhouse gas emissions, and enhance the resilience of urban environments against climate change (Eveloy and Ayou, 2019). Innovations in fifth-generation district heating and cooling (5GDHC) systems, which operate at near-ambient temperatures and incorporate renewable energy sources, have shown potential, though implementation challenges persist (Gjoka et al., 2023).

Furthermore, the relationship between building characteristics and cooling demand has been widely studied, underscoring the influence of insulation, glazing, thermal performance, and operational strategies (Bansal and Quan, 2022; Mutani et al., 2022). Studies on various buildings, have demonstrated that inefficient HVAC operation and poor insulation significantly increase energy consumption, while passive cooling techniques and smart controls mitigate demand (Asim et al., 2022; A.B.N.A.A, 2021). Beyond building characteristics, cooling load is also driven by natural factors (temperature, humidity, ventilation, solar radiation) and social factors (policies, technology, human behavior) (a et al., 2021). The Environment-People-Building framework provides an interdisciplinary approach to assessing district cooling efficiency. Building morphology, including floor number, exposed surface area, and conditioned zones, also plays a crucial role in cooling demand (Ravichandran and Gopalakrishnan, 2024). Moreover, uncertainty factors such as occupancy behavior and energy system variability further complicate cooling load prediction. Studies on Variable Refrigerant Flow (VRF) systems in Middle Eastern climates demonstrate that cooling set points and HVAC usage per area significantly impact energy consumption, with greater uncertainty effects in hyper-arid regions (Saryazdi et al., 2024).

In recent years, machine learning has emerged as a powerful tool for enhancing cooling load prediction by capturing complex, nonlinear relationships between building parameters and energy demand (Lu et al., 2023). For instance, studies have demonstrated the efficacy of support vector regression (SVR) in optimizing air conditioning system operations (Huang et al., 2022) and evaluated multiple ML techniques, such as decision trees, neural networks, and logistic regression, for residential cooling load forecasting. One such comparative study found that a decision tree model achieved a high accuracy of 93.24 % for cooling load, while a linear regression model achieved an R^2 of 0.888 (Mehdizadeh Khorrami et al., 2024). Building on this, advanced deep-learning models, such as aTCN-LSTM, have integrated multi-head temporal convolutional networks and BiLSTM to improve prediction accuracy, achieving up to a 27.48 % reduction in mean absolute percentage error (MAPE) and a single-step R^2 of 0.971 (Song et al., 2024). Similarly, hybrid models, such as the RF-IPWOA-ELM which combines Random Forest (RF) for feature selection, an Improved Parallel Whale Optimization Algorithm (IPWOA) for parameter tuning, and an Extreme Learning Machine (ELM) for forecasting, have demonstrated enhanced efficiency in commercial buildings, with a reported MAPE as low as 0.2 % (Gao et al., 2022).

More specifically, short-term cooling load forecasting in Building District Energy Systems (BDESs) has significantly improved with the use of machine learning models such as XGBoost, LSTM, and attention-based frameworks (Yu et al., 2023). In one such study, XGBoost was found to be the best-performing model for one-hour-ahead forecasts, achieving a CV-RMSE of 14.51 % for cooling loads. In hot and humid climates, machine learning meta-models have demonstrated promising results in predicting cooling loads for high-rise residential district cooling systems (Jia et al., 2022). For example, Jia et al. (2022) developed an ANN-based meta-model using simulated data that reported a test R^2 of 0.9991 and a CV(RMSE) of 1.9 % (Jia et al., 2022). Despite these advancements, most existing research has focused on district heating rather than cooling, highlighting a gap in ML applications tailored to DCS (Ntakolia et al., 2022).This underscores the need for more data-driven approaches specific to cooling load optimization and energy efficiency improvements.

In parallel with predictive modeling, the integration of renewable energy into DCS has gained increasing attention as a means to enhance energy efficiency and reduce carbon emissions. Inayat and Raza (2019) explored various renewable sources, including biomass, solar thermal, geothermal, surface water, and waste heat energy, as viable alternatives for powering DCS. Their findings highlight that geothermal-assisted cooling and biomass-powered DCS can significantly cut CO_2 emissions while utilizing waste materials, making them economically feasible for urban applications. Among the various renewable options, solar energy has been identified as a particularly promising option for district cooling, especially in hot climates. Ismaen et al. (2022) analyzed different solar-assisted DCS configurations, including Photovoltaic (PV), Photovoltaic-Thermal (PVT), and Thermal (T) solar systems. Their study found that PV-DCS can reduce environmental impact by 58.3 %, while PVT-DCS achieves the highest environmental performance, decreasing global warming potential by 89.5 %. However, economic feasibility is highly dependent on electricity tariffs and available installation areas, with higher subsidies favoring grid-powered DCS over renewable integration

While these findings highlight the theoretical potential of solar-assisted DCS, most studies are based on simulations or financial models. There remains a lack of empirical analysis evaluating the real-world carbon reduction impact of solar integration at varying penetration levels within operational DCS networks.

2.1. Research gaps and contributions

While foundational, existing research on cooling load forecasting predominantly relies on simulated data or case studies of single buildings. These approaches inherently fail to capture the operational complexities and occupant-driven variability across a diverse building stock, limiting their real-world accuracy and generalizability. Furthermore, while individual components like explainable AI or solar assessments exist, they are often studied in isolation. There is a distinct lack of integrated frameworks that bridge the gap from data-driven prediction to actionable, evidence-based renewable energy planning for an entire district.

2.2. This study directly addresses these critical limitations. Our primary contributions are

- A Unique District-Level Dataset: We analyze a real-world operational dataset from a cohort of 59 residential buildings serviced by a single district cooling system (DCS). This provides a rare, empirical view into a complete urban energy system, allowing us to develop a generalizable model that accounts for the real-world diversity absent from simulation or single-building studies.
- An Integrated, Interpretable Framework: We introduce a novel machine learning pipeline that not only predicts cooling load with high accuracy (Random Forest) but also ensures transparency by identifying key demand drivers through explainable AI (SHAP). This moves beyond "black-box" models to support trusted, data-driven decision-making.
- Evidence-Based Renewable Energy Planning: We conduct a comprehensive solar integration assessment to quantify the carbon reduction potential of varying PV penetration levels (25 %, 50 %, and 75 %). Uniquely, this analysis is grounded in the empirically derived, real-world cooling load profiles of the entire district, not on theoretical assumptions.
- A Focus on an Underrepresented Domain: Our research provides crucial insights into the residential district cooling sector in a hot, arid climate—a context of growing global importance that remains significantly understudied in ML-based energy research.

The following section details the methodology employed in this research, including data collection, preprocessing, feature selection, model development, and validation.

3. Methods

3.1. Data collection and preprocessing

The foundation for this study is a unique, real-world dataset integrating three primary sources: district cooling consumption records, building characteristics, and meteorological data.

The core consumption data, provided by the district cooling utility, consists of monthly cooling energy consumption (measured in refrigeration ton-hours, TRH) for 59 residential buildings in Dubai, UAE. This building stock was commissioned around 2008 and the data spans a four-year period from November 2020 to October 2024. The final dataset contains 2781 complete records, representing a data availability of 98.2 % after accounting for occasional metering gaps. To account for architectural diversity, each building was categorized into one of four typologies (designated A, B, C, and D) based on its total built-up area and building configuration. All building-related data, including floor area, was converted to SI units for consistency.

Table 1 details the key characteristics of these building typologies. Additionally, occupancy rates are included to account for variations in usage intensity. The analysis was conducted at a monthly temporal resolution, as this aligned with the granularity of the billing data provided by the utility.

Meteorological data were sourced from the ECMWF ERA5 reanalysis dataset (C.C.C.S. C3S,), pre-aggregated to a monthly resolution that aligns with the temporal scale of the consumption data. The selected variables included mean monthly temperature (t2m in °C), mean monthly relative humidity (RH in %), and total monthly solar radiation (ssrd in MJ/m^2). After integrating all data sources, the dataset was structured for predictive modeling by defining the Cooling Load (TRH) as the target variable. The input features were organized into numerical (Floor Area, Occupancy, Temperature, Relative Humidity, Solar Radiation, Month) and categorical (Building Number, Building Typology) groups. The 'Building Number' and 'Building Typology' features were included as categorical variables and processed using one-hot encoding. This allows the model to learn the unique, static properties of each building, such as unobserved differences in envelope or HVAC efficiency.

A systematic preprocessing pipeline was implemented using Scikit-learn's ColumnTransformer to prepare the features for machine learning. This approach ensures all data transformations are consistently applied, preventing data leakage and enhancing reproducibility. The pipeline first handled missing values in the dataset and subsequently, all numerical features were standardized using StandardScaler to have a mean of zero and a standard deviation of one, while categorical features were converted into a numerical format using OneHotEncoder with sparse_output=False to produce a dense array. This structured process resulted in a fully processed dataset ready for model training and validation.

3.2. Exploratory data analysis (EDA)

A comprehensive exploratory data analysis (EDA) was conducted to dissect the relationships between building characteristics, meteorological conditions, and cooling energy consumption.

Initial analysis focused on macro-level trends. A Pearson correlation

Table 1
Characteristics of building typologies.

Building Typology	Floor Area, m^2	Number of Buildings	Building Configuration
A Type	6382.4	28	G+ 4
B Type	7528.9	16	G+ 5
C Type	7635.2	10	G+ 5
D Type	8936.2	5	G+ 6

matrix was generated to quantify the linear associations between the cooling load and all other variables. Pearson correlation was selected as a preliminary filter to identify linear trends; we acknowledge this method does not capture non-linearities, which are instead handled by the subsequent tree-based machine learning models. Concurrently, time series plots of the aggregate cooling demand revealed long-term consumption trends, while monthly boxplots provided insights into seasonal fluctuations, distribution (min, median, max), and potential outliers.

To understand the influence of architectural design, the average monthly cooling load for each building typology (A, B, C, and D) was analyzed. To enable a standardized comparison of energy efficiency independent of building size or population, this analysis was extended by normalizing consumption in two ways: by floor area to calculate cooling load intensity (TRH/m^2) and by occupant count to derive cooling load per occupant (TRH/occupant). These normalized metrics were then visualized to identify whether specific typologies consistently demonstrated higher or lower energy performance.

Finally, to investigate the impact of space utilization, the relationship between occupant density (occupants/m^2) and cooling load intensity was examined using scatter plots. This helped determine if buildings with denser occupancy exhibited disproportionately high cooling loads. The insights derived from this multi-faceted EDA formed the empirical basis for the feature engineering and predictive modeling detailed in subsequent sections.

3.3. Predictive modeling of cooling load

This section details the predictive models developed to forecast the monthly cooling load. To identify the most accurate approach, a comparative analysis was conducted using four machine learning algorithms, selected to span a range of complexities from a simple linear baseline to advanced ensembles. Linear Regression was employed as a fundamental baseline to establish a performance benchmark. To capture complex non-linear interactions, a Random Forest Regressor was utilized for its known robustness. Additionally, two powerful sequential ensemble algorithms were evaluated: the standard Gradient Boosting Regressor and the XGBoost Regressor. The latter was selected as it represents a regularized and computationally efficient implementation of gradient boosting, renowned for its state-of-the-art performance on tabular data.

To ensure a robust, real-world evaluation and prevent data leakage, a strict chronological train-test split was employed. All data from November 2020 to July 2023 (1914 records) was used for the training set, and all data from August 2023 to October 2024 (867 records) was held out as the final, unseen test set.

3.3.1. Linear regression (Baseline Model)

Linear Regression (LR) is a fundamental statistical method used for modeling the linear relationship between a dependent variable and one or more independent variables. As described in (Hastie et al., 2009), the linear regression model assumes that the relationship between these variables can be approximated by Eq. (1):

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 \ldots\ldots\ldots\ldots + \beta_n X_n + \in \tag{1}$$

where Y is the predicted cooling load, $X_1, X_2, \ldots, X_n$ are the independent variables, β_0 is the intercept, $\beta_1, \beta_2, \ldots, \beta_n$ are the coefficients representing the influence of each independent variable on the cooling load, and ε is the error term, representing the unexplained variance.

The coefficients $\beta_0, \beta_1 \ldots, \beta_n$ are estimated using the method of least squares. Given a set of N training observations (x_i, y_i), where $x_i = (x_{i1}, x_{i2}, \ldots, x_{in})$ represents the predictor variables for the i^{th} observation and y_i represents the corresponding cooling load, the least squares method aims to minimize the residual sum of squares (RSS) given by Eq. (2).

$$RSS(\beta) = \sum_i \left(y_i - \beta_0 - \sum_j \beta_j x_{ij} \right)^2 \tag{2}$$

This minimization problem yields the optimal coefficients that best fit the linear relationship to the observed data.

The coefficients β_j provide valuable insights into the relationship between the predictors and cooling load. However, Linear Regression relies on the assumption of a linear relationship, which may not always hold in complex systems like DCS. Additionally, it assumes that the errors are independent and identically distributed, which may not be the case in real-world scenarios. Despite these limitations, Linear Regression serves as a useful baseline model due to its simplicity and interpretability. It provides a benchmark against which the performance of more complex non-linear models can be compared.

3.3.2. Random forest

Random Forest (RF) is an ensemble learning method that constructs multiple decision trees during training and outputs the mean prediction of the individual trees for regression tasks. As (Breiman, 2001) describes, the algorithm operates by creating a multitude of decision trees at training time and outputting the mean/average prediction of the trees for regression. Each tree is constructed through a process involving bootstrapping, where a random sample with replacement is drawn from the original training data to create a new training set, and feature randomness, where at each node of a tree, instead of considering all possible features for splitting, a random subset of features is considered, with the best split among these features being used to grow the tree. These two sources of randomness contribute to the diversity of the individual trees, making them less correlated and the ensemble prediction more robust. For regression tasks, the Random Forest Regressor combines the predictions of individual trees by averaging them. Given a new input x, each tree in the forest produces a prediction $T_i(x)$, and the final prediction of the Random Forest is calculated as the average of these individual predictions, as shown in the Eq. (3).

$$\hat{y}(x) = \left(\frac{1}{B}\right) * \sum_i T_i(x) \tag{3}$$

where B is the number of trees in the forest. Random Forest offers several advantages, including its ability to handle non-linear relationships between predictors and the outcome variable without requiring explicit transformations of the input features, its reduction of overfitting due to the ensemble nature and randomness in tree construction, its robustness to outliers because of the use of bootstrapping, and its provision of a measure of feature importance, indicating which predictors have the strongest influence on the outcome. However, Random Forest also has limitations, such as the complexity of interpretation due to the large number of trees and the potential for high computational cost, especially for very large datasets.

3.3.3. Gradient boosting regressor

Gradient Boosting (GB) is another ensemble method that builds trees sequentially, with each tree correcting the errors of its predecessors (Friedman, 2001). It works by iteratively adding new trees to the ensemble, where each new tree h(x; γ) is trained to minimize the residual errors of the previous trees. Specifically, given the current ensemble of m trees $f_{m-1}(x)$, the next tree is fit to the negative gradient of the loss function L(y, f(x)) with respect to the predictions f(x), evaluated at the current ensemble's predictions in Eq. (4)

$$g_m = -\frac{\partial L(y, f(x))}{\partial f(x)}\bigg|_{\{f(x) = f_{(m-1)}(x)\}} \tag{4}$$

This gradient g_m represents the direction of steepest descent in function space, and the new tree h(x; γ) is fit to approximate this negative gradient. The parameter γ represents the parameters of the tree (e.g., split points, leaf values). This process can be understood as a functional gradient descent, where we are iteratively updating our function estimate f(x) by moving in the direction of the negative gradient g_m. We then scale the contribution of the new tree $h(x; \gamma_m)$ by a learning rate ν to control the step size of this update. The update step for the ensemble is then shown in Eq. (5)

$$f_{m(x)} = f_{\{m-1\}(x)} + \nu * h(x; \gamma_m) \tag{5}$$

By iteratively adding trees in this manner, Gradient Boosting effectively minimizes the loss function L(y, f(x)). In the context of DCS cooling load prediction, Gradient Boosting can effectively model the time-dependent nature of cooling demand and the influence of factors like weather forecasts and building usage patterns. Similar to Random Forest, this model incorporates a wider range of features, to capture temporal dependencies. Gradient Boosting is known for its high predictive accuracy and ability to handle complex relationships, making it suitable for the intricacies of DCS cooling load forecasting.

3.3.4. XGBoost regressor

XGBoost (XGB), is a sophisticated machine learning algorithm that falls under the category of ensemble learning methods, specifically a refined implementation of Gradient Boosting (Chen and Guestrin, 2016). Gradient Boosting combines multiple "weak learners," typically decision trees, to create a strong predictive model, with each tree trained to correct the errors of its predecessors. XGBoost distinguishes itself through several key features: regularization (L1 and L2) to prevent overfitting, the ability to handle missing data by learning optimal imputation strategies during training, and parallel processing for faster computation. These features are particularly valuable in the context of DCS cooling load prediction, where data from building management systems and weather stations may be noisy or incomplete, and rapid model training is crucial for real-time applications. XGBoost's objective function at iteration t is defined as in Eq. (6).

$$L(t) = \sum_i l\left(y_i, \widehat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t) \tag{6}$$

where L(t) is the objective at iteration t, $l(y_i, \hat{y}_i^{(t-1)})$ is the loss function, $f_t(x_i)$ is the new tree, and $\Omega(f_t)$ is the regularization term.This objective, optimized via a second-order Taylor approximation and gradient descent, allows XGBoost to efficiently determine the best tree to add at each step. The combination of these features makes XGBoost a powerful tool for various machine learning tasks, including the complex challenge of forecasting cooling loads in dynamic urban environments.

3.3.5. Model evaluation metrics

To assess the performance of the predictive models developed in this study, a suite of evaluation metrics was employed. These metrics provide complementary perspectives on the accuracy and goodness-of-fit of the models, enabling a comprehensive comparison of their predictive capabilities. The following metrics were specifically utilized:

Mean Squared Error (MSE): The MSE quantifies the average squared difference between the predicted and actual cooling load values given by Eq. (7).

$$MSE = \left(\frac{1}{n}\right) * \sum_i (y_i - \widehat{y}_i)^2 \tag{7}$$

where n is the number of data points, y_i represents the actual cooling load for the i^{-th} data point, and $\hat{y}_i$ represents the predicted cooling load for the i^{-th} data point. MSE is sensitive to outliers due to the squaring of the errors, giving higher weight to larger prediction errors. It is a commonly used metric for evaluating regression models, providing a measure of the overall prediction accuracy. MSE is a standard metric for assessing the average squared difference between predictions and actual values.

Mean Absolute Error (MAE): The MAE measures the average absolute difference between the predicted and actual cooling load values. It is calculated as per Equation (8).

$$MAE = \left(\frac{1}{n}\right) * \sum_i |y_i - \widehat{y}_i| \tag{8}$$

where n, y_i, and $\hat{y}_i$ are defined as above. MAE is less sensitive to outliers compared to MSE, as it uses absolute differences rather than squared differences. It provides a more direct interpretation of the average prediction error in the original units of the cooling load. Similar to MSE, MAE is a widely used metric in regression, offering a measure of prediction accuracy that is less influenced by extreme values.

R-squared (R^2): The R-squared coefficient represents the proportion of the variance in the dependent variable (cooling load) that is predictable from the independent variables which is explained in Eq. (9).

$$R^2 = 1 - \left(\frac{\sum_i (y_i - \widehat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}\right) \tag{9}$$

where n, y_i, and $\hat{y}_i$ are defined as above, and $\bar{y}$ represents the average of the actual cooling load values. R^2 ranges from 0 to 1, with higher values indicating a better fit. An R^2 of 1 implies a perfect fit, while an R^2 of 0 suggests that the model does not explain any variance in the cooling load. R^2 provides a measure of how well the model generalizes to unseen data, with higher values indicating better generalization. R^2 quantifies the proportion of variance in the target variable explained by the model, providing insight into the goodness-of-fit.

Adjusted R-squared (Adjusted R^2): To account for the number of predictors in the model and avoid overestimating the goodness-of-fit, we also utilize Adjusted R-squared. Adjusted R^2 penalizes the inclusion of unnecessary predictors, providing a more reliable measure of model fit, particularly when comparing models with different numbers of independent variables. It is calculated as per Eq. (10).

$$AdjustedR^2 = 1 - \left[\frac{(1 - R^2) * (n-1)}{(n-k-1)}\right] \tag{10}$$

where n is the number of observations and k is the number of predictors. Higher Adjusted R^2 values indicate a better fit, adjusted for the number of predictors, and help prevent overfitting. Like R^2, Adjusted R^2 provides insight into model generalization.

Root Mean Squared Error (RMSE): RMSE measures the average magnitude of prediction errors in a regression model, providing an estimate of how closely predicted values align with actual outcomes. It is calculated as in Eq. (11).

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \widehat{y}_i)^2} \tag{11}$$

where n is the number of data points, y_i represents the actual value for the i^{-th} data point, and $\hat{y}_i$ represents the predicted value for the i^{-th} data point. A lower RMSE indicates a better fit of the model to the data, signifying more accurate predictions. RMSE is particularly useful when large errors are especially undesirable, as it penalizes them more heavily due to the squaring of each error term. According to (Hodson, 2022), RMSE is optimal for normally distributed errors, while MAE is more suitable for Laplacian error distributions.

The combined use of these metrics provides a robust evaluation of the predictive models, considering both the magnitude of prediction errors (RMSE, MSE and MAE) and the proportion of variance explained by the model (R^2 and Adjusted R^2). This comprehensive evaluation approach allows for a thorough comparison of the models and the selection of the most suitable model for cooling load prediction in district cooling systems. These metrics are widely used in regression analysis and are discussed in detail in standard statistical learning texts (James et al., 2023).

3.3.6. Hyperparameter tuning

To ensure the most optimal model was selected, the performance of the default models was first benchmarked. The best-performing baseline model was then subjected to a systematic hyperparameter tuning process to determine if its performance could be further improved. We employed Optuna, an automated hyperparameter optimization framework (Akiba et al., 2019), to manage this process. The optimization objective was to minimize the Mean Squared Error (MSE) on a dedicated validation set (a chronological subset of the training data) over 50 trials. The hyperparameters tuned for the RF model included n_estimators, max_depth, min_samples_split, and min_samples_leaf. The results of this tuning and the final model selection are detailed in Section 4.3.

3.4. Feature importance analysis using SHAP

To evaluate the factors influencing district cooling load predictions, we conducted a feature importance analysis using the model and SHAP values. This analysis quantifies the contribution of key predictors, including floor area, meteorological variables, and occupancy, in driving cooling energy demand.

SHAP values further enhance interpretability by illustrating how each feature impacts the model's predictions. As described in (Lundberg and Lee, 2017), SHAP, derived from cooperative game theory, explains the contribution of each feature to individual predictions by distributing the total model output across input variables. SHAP values offer a unified framework for interpreting model predictions by assigning each feature an importance value for a particular prediction. This approach is particularly valuable for complex models like RF, where understanding the specific influence of each feature can be challenging. Mathematically, the SHAP value for a feature i is given by Eq. (12)

$$\phi_i = \sum_{S\subseteq F\setminus\{i\}} \frac{|S|!.(|F|-|S|-1)!}{|F|!}[f(S\cup\{i\})-f(S)] \tag{12}$$

The Shapley value (ϕ_i), for each feature i was calculated by considering all feature subsets S (excluding i) and their corresponding model predictions f(S). This method provides a fair assessment of each feature's contribution. The SHAP explainer generated two key visualizations: a bar plot of mean absolute SHAP values, indicating overall feature importance, and a SHAP summary dot plot, showing the distribution of individual prediction SHAP values across different scenarios. These visualizations offer a comprehensive understanding of how building characteristics and external factors influence cooling energy demand, providing valuable insights for energy-efficient design optimization, as advocated by (Lundberg and Lee, 2017) for a more unified and informative interpretation of complex model predictions beyond simple feature rankings.

3.5. Estimating carbon emissions from district cooling consumption

This section details the methodology used to estimate carbon emissions from district cooling consumption in the analyzed residential buildings. The approach involves converting predicted cooling loads into electricity consumption, applying UAE-specific emission factors that consider ongoing grid decarbonization, and analyzing scenarios to evaluate potential emissions reduction.

3.5.1. Baseline carbon emission calculation

Baseline carbon emissions were calculated through a two-step process. First, to account for the UAE's ongoing grid decarbonization, annual emission factors (EF) were sourced from the CaDI database (CaDI, 2024). These factors, measured in $kgCO_2e/kWh$ (kilograms of carbon dioxide equivalent per kilowatt-hour), reflect the evolving energy generation mix for each year of the study period (Table 2). The use of $kgCO_2e$ standardizes the warming potential of all greenhouse gases relative to CO_2, ensuring consistent accounting. Second, the predicted cooling load, measured in refrigeration ton-hours (TRH), was converted to electricity consumption in kilowatt-hours (kWh). This conversion was performed using an established chiller plant efficiency factor of 0.92 kW/TR, a standard benchmark for systems in the region as documented in regional studies (President et al., 2016). This value represents an annual average plant efficiency and is a simplifying assumption, as detailed plant-level data on part-load performance, seasonality, or distribution losses was not available. The direct conversion is shown in Eq. (13):

Table 2
UAE annual emission factors for electricity generation.

Year	Emission Factor ($kgCO_2e/kWh$)
2020	0.49353
2021	0.48476
2022	0.48437
2023	0.45295
2024	0.42161

Source: (CaDI, 2024).

$$E_{consumption}(\text{kWh}) = \text{Cooling Load}(\text{TRH}) \times 0.92 \tag{13}$$

The building's total baseline emissions were then calculated by multiplying its annual electricity consumption by the corresponding year's emission factor.

3.5.2. Energy efficiency scenarios

To evaluate the impact of energy efficiency improvements on emissions, three scenarios were modeled representing 10 %, 20 %, and 30 % reductions in the predicted cooling-related electricity consumption.

The reduced energy use ($E_{reduced}$), corresponding carbon emissions ($CO_{2\ reduced}$), and subsequent emission savings ($CO_{2\ savings}$) for each scenario were calculated sequentially using Eqs. (14) through (16).

$$E_{reduced} = E_{consumption} * (1 - Reduction\ Rate) \tag{14}$$

$$CO_{2reduced} = E_{reduced} * EF_{year} \tag{15}$$

$$CO_{2saving} = CO_{2original} - CO_{2reduced} \tag{16}$$

In these equations, $E_{consumption}$ represents the baseline electricity consumption (kWh); Reduction Rate is the targeted efficiency improvement (0.10, 0.20, or 0.30); EF_{year} is the specific emission factor for the corresponding year ($kgCO_2e/kWh$); and $CO_{2original}$ is the baseline carbon emissions before applying efficiency measures ($kgCO_2e$).

This analysis assumes a direct, linear relationship between electricity consumption and carbon emissions which is a standard and consistent practice in carbon accounting methodologies (Schmitz et al., 2004).

3.5.3. Solar energy integration scenarios

To model the impact of on-site renewable energy, three solar energy integration scenarios were developed. These scenarios represent solar photovoltaic (PV) generation offsetting 25 %, 50 %, and 75 % of the cooling-related electricity demand.

The resulting net carbon emissions after accounting for the solar offset (CO_2solar) were calculated for each scenario using Eq. (17), as described by Schmitz et al. (2004).

$$CO_{2solar} = CO_{2original} * (1 - solar_{offset}) \tag{17}$$

In this equation, $CO_{2\ solar}$ represents the carbon emissions after accounting for the solar offset ($kgCO_2e$), $CO_{2\ original}$ is the baseline carbon emissions before solar integration ($kgCO_2e$), and solar_offset is the proportion of electricity demand met by solar PV, expressed as a decimal (0.25, 0.50, or 0.75).

Finally, the emissions calculated under each energy efficiency and solar integration scenario were aggregated and compared to evaluate the most effective carbon mitigation pathways.

4. Results

4.1. Correlation analysis

To quantify the linear relationships between the numerical variables in the dataset, a Pearson correlation analysis was conducted. The resulting correlation matrix is presented in Fig. 1.

The analysis reveals a strong positive correlation (r = 0.79) between cooling load and outdoor temperature, identifying it as the primary driver of energy consumption. The analysis also showed a weak positive correlation between cooling load and both Occupancy (r = 0.23) and Total BUA (r = 0.16). Conversely, a moderate negative correlation was observed between cooling load and relative humidity (r = -0.30), while total monthly solar radiation (ssrd) showed only a weak positive correlation (r = 0.29).

Significant inter-correlations were also found among the meteorological variables themselves. Notably, there was a strong positive relationship between temperature and solar radiation (r = 0.69) and a strong negative relationship between relative humidity and solar radiation (r = -0.73), highlighting the complex interplay of climatic factors.

4.2. Seasonal cooling load analysis based on typology, area, and occupancy

This section analyzes cooling load characteristics across the four residential building typologies (A, B, C, and D) by examining total consumption, consumption intensity per unit area, and consumption per occupant.

As shown in Fig. 2a, a consistent seasonal pattern was observed across all typologies, with demand peaking in August. The boxplot format visualizes the full distribution (including minimum, median, and maximum values) of the loads, and the overlaid dual-axis plot (red line) confirms the strong correlation between cooling load and the average monthly temperature. In terms of total consumption, D-type buildings, which have the largest average floor area, exhibited the highest overall cooling load throughout the year.

When normalized by floor area (Fig. 2b), the results for cooling intensity (TRH/m^2) were inverted. B-type buildings demonstrated the highest cooling intensity, while D-type buildings, despite their size, showed the lowest, indicating greater efficiency per unit area. This divergence in cooling intensity among typologies was most pronounced during the peak summer months.

Analysis of cooling load per occupant (Fig. 2c) revealed that D-type buildings also had the highest consumption on a per-person basis. Significant monthly variability was noted in A and B-type buildings; however, statistical checks confirmed these were not data outliers. Specifically, a Z-score analysis of the entire dataset confirmed that only 4 of the 2781 data points (0.14 % of the dataset) fell outside a 3-sigma threshold, and these were retained as genuine operational fluctuations.

Finally, the relationship between occupant density (occupants/m^2)

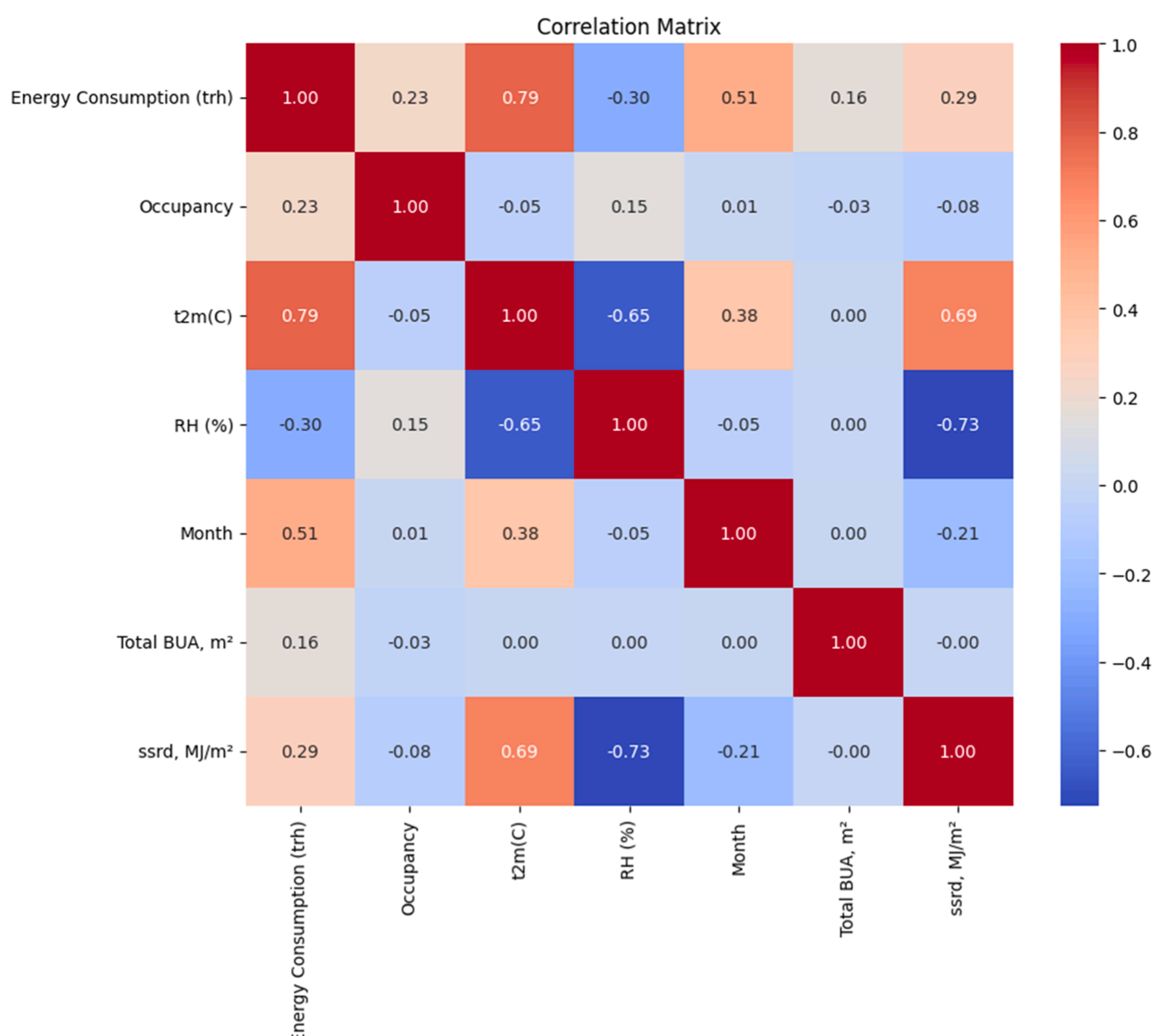

Fig. 1. Pearson correlation matrix of numerical variables.

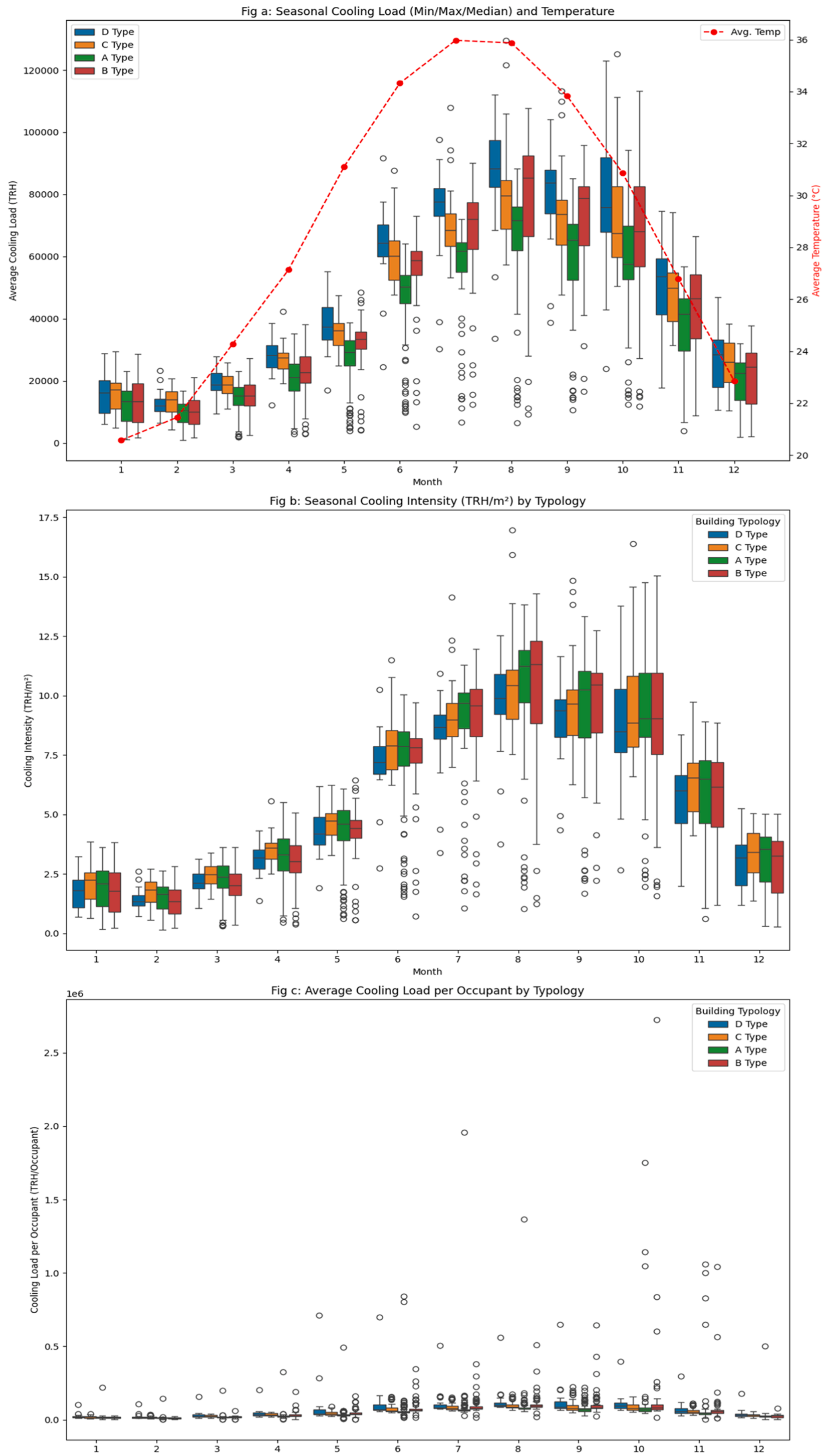

Fig. 2. Combined seasonal analysis: (a) Seasonal cooling load (Min/Max/Median) and temperature, (b) Seasonal cooling intensity (TRH/m^2) by Typology, (c) Average cooling load per occupant by typology.

and cooling intensity (TRH/m^2) was examined (Fig. 3). A complex, non-linear relationship was observed. Notably, some high-density buildings (e.g., B and C types) achieved low cooling intensities, while the lower-density D-type buildings often exhibited higher intensities. This identifies occupant density as a key, non-linear predictor of cooling performance.

4.3. Forecasting model performance

To evaluate the efficacy of different machine learning algorithms in predicting district cooling energy consumption, four regression models (LR, RF, GB, and XGB) were developed. The models were assessed using Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2, and Adjusted R^2.

4.3.1. Model benchmarking and selection

The four models were trained and then evaluated on the strictly chronological test set defined in Section 3.3 (all 867 records from August 2023 to October 2024). The performance of the models on this held-out data is summarized in Table 3.

The results demonstrate that the RF model provided the best performance, emerging as the most effective model. It yielded the highest R^2 (0.8256) and Adjusted R^2 (0.8105), as well as the lowest RMSE (11,668.31). This indicates that the RF model could explain 82.56 % of the variance in future cooling load, significantly outperforming the other ensemble methods in this time-aware test.

To further validate the stability of this result, a 1000-iteration bootstrap analysis was conducted. This confirmed the robustness of the RF model, yielding a 95 % confidence interval of [0.8050, 0.8443] for the $R^2$and [119,129,423.49, 153,713,354.13] for the MSE.

4.3.2. Hyperparameter optimization and overfitting assessment

To ensure the most optimal model was used, the best-performing model from the benchmark, RF model, was subjected to hyperparameter tuning using Optuna. This tuning process, which optimized n_estimators, max_depth, min_samples_split, and min_samples_leaf, resulted in a model with a lower performance on the final, unseen test set (R^2=0.8158) compared to the default model (R^2=0.8256). This indicates that the default parameters provided a more robust and generalizable model, while the tuned parameters were slightly overfit to the validation subset.

Table 3
Model performance on chronological test data.

Model	MSE	MAE	R^2	Adjusted R^2	RMSE
LR	198,000,105.02	10,475.12	0.7464	0.7245	14,071.25
RF	136,149,571.95	8786.74	**0.8256**	0.8105	11,668.31
GB	176,168,100.34	9520.48	0.7744	0.7549	13,272.83
XGB	176,764,846.35	9820.57	0.7736	0.7540	13,295.29

4.3.3. Final model selection

Based on this comprehensive validation and robustness check, the default RF model was selected as the final, definitive model for all subsequent SHAP interpretability and scenario analyses. Its robust performance on the unseen chronological test set was:

- **R^2 :** 0.8256 (95 % CI: **[0.8050, 0.8443]**)
- **Adjusted R^2 :** 0.8105
- **MSE:** 136,149,571.95 (95 % CI: **[119,129,423.49, 153,713,354.13]**)
- **RMSE:** 11,668.31 (95 % CI: **[10,914.64, 12,398.12]**)
- **MAE:** 8786.74

4.4. Model interpretation with SHAP

To interpret feature contributions to the cooling load predictions, a SHAP analysis was conducted on the final validated Random Forest model. The SHAP feature importance plot (Fig. 4) displays the mean absolute SHAP value for each feature, representing its overall impact on the model's output. Temperature (t2m(C)) emerged as the most influential predictor by a significant margin, aligning with the strong correlation observed in the EDA. This is followed by Month and Occupancy, which have a similarly strong impact. The remaining features, ranked in

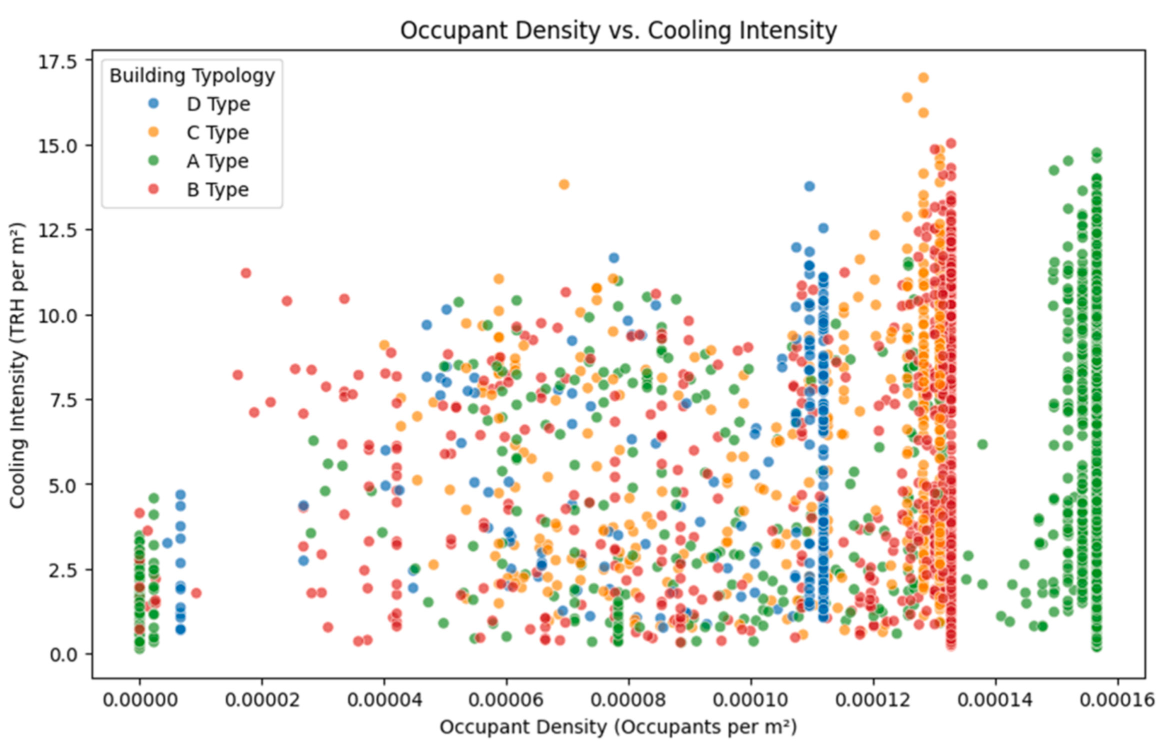

Fig. 3. Relationship between occupant density and cooling intensity (TRH/m^2).

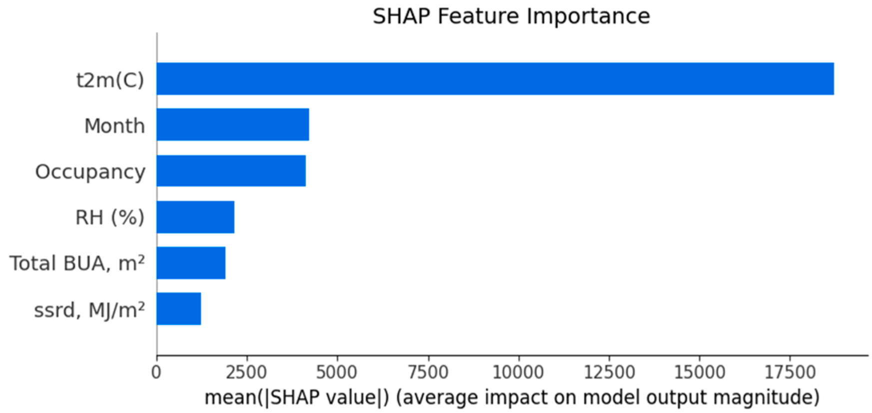

Fig. 4. SHAP feature importance plot.

order of importance, were Relative Humidity (RH (%)), Total BUA (m^2), and Solar Radiation (ssrd, MJ/m^2).

The SHAP summary plot (Fig. 5) clarifies the directional impact of these features. As expected, higher values of Temperature, Occupancy, and Total BUA are all associated with increased cooling load predictions (positive SHAP values). The plot for Month reveals a clear non-linear seasonal pattern: low and high feature values (representing non-summer months) correspond to negative SHAP values, while intermediate values (representing the summer season) drive the load higher. The model also identified a positive relationship for Relative Humidity, where high humidity is generally associated with a higher cooling load (positive SHAP values), and low humidity is associated with a lower load. This SHAP analysis validates the predictive relevance of the selected features and confirms that the final model's decision-making process is interpretable and aligns with physical building energy principles.

4.5. Impact of energy efficiency scenarios on carbon emissions

The projected impact of applying 10 %, 20 %, and 30 % energy efficiency (EE) improvements on carbon emissions is illustrated in Fig. 6. The baseline (0 % reduction) emissions trend upwards from 2020 to 2023, followed by a decrease in 2024 that reflects the grid's improving carbon intensity. Each EE scenario yields a proportional reduction in emissions relative to this baseline. The magnitude of this impact is substantial; for example, in the peak year of 2023, the 30 % efficiency scenario results in an absolute carbon reduction of approximately 3.7×10^6 $kgCO_2e$ compared to the baseline

4.6. Comparative analysis with solar energy integration

The impact of solar integration is compared to the energy efficiency pathways in Fig. 7. The analysis reveals that solar offsets provide a more potent mitigation strategy than demand-side reductions. A key finding is that the 25 % solar offset scenario surpassed the savings of the most aggressive 30 % energy efficiency scenario. This comparison, however, rests on annual-energy calculations and does not account for temporal matching between generation and demand. The 75 % solar offset scenario offered the maximum mitigation potential, lowering the 2023 baseline emissions by an estimated 8.9×10^6 $kgCO_2e$.

5. Conclusion and discussion

This study introduced and validated a novel framework integrating machine learning with scenario analysis to optimize district cooling systems and reduce associated carbon emissions in the UAE. By leveraging a comprehensive real-world dataset, this research successfully developed a forecasting model for cooling load and evaluated the impact of various interventions on carbon reduction. The findings confirm the significant influence of temperature, occupancy patterns, and solar radiation on cooling load. The Random Forest model demonstrated high predictive accuracy, while the SHAP analysis provided valuable insights into the key drivers of cooling demand. Scenario analysis revealed that both energy efficiency measures and solar energy integration can substantially reduce carbon emissions, with even moderate improvements yielding notable carbon savings and high solar penetration demonstrating the potential to achieve near-zero emissions

A thorough understanding of these findings offers several critical implications. The strong positive correlation between temperature and

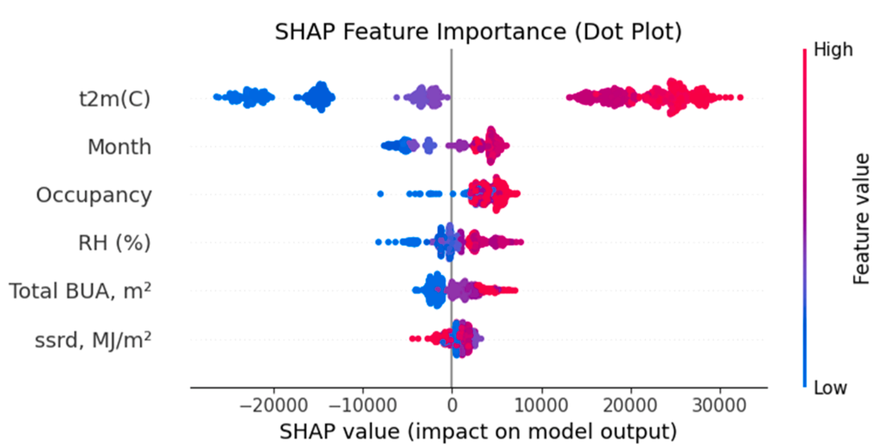

Fig. 5. SHAP summary plot.

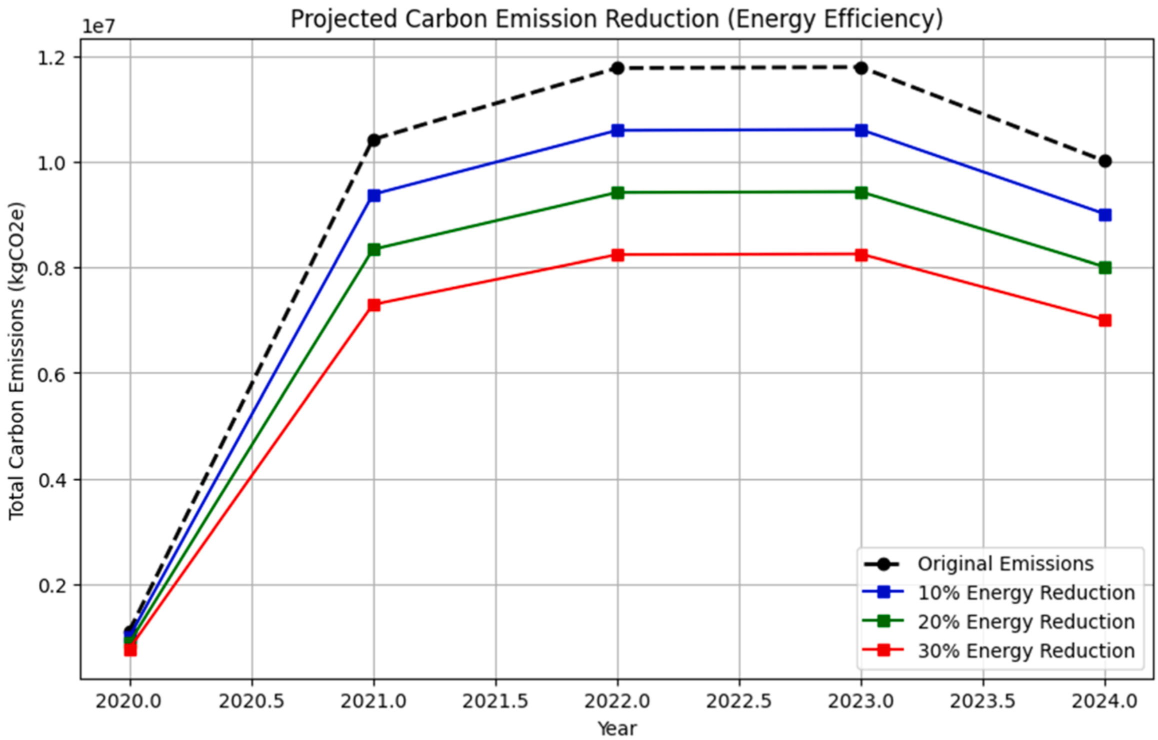

Fig. 6. Carbon emission reduction scenarios.

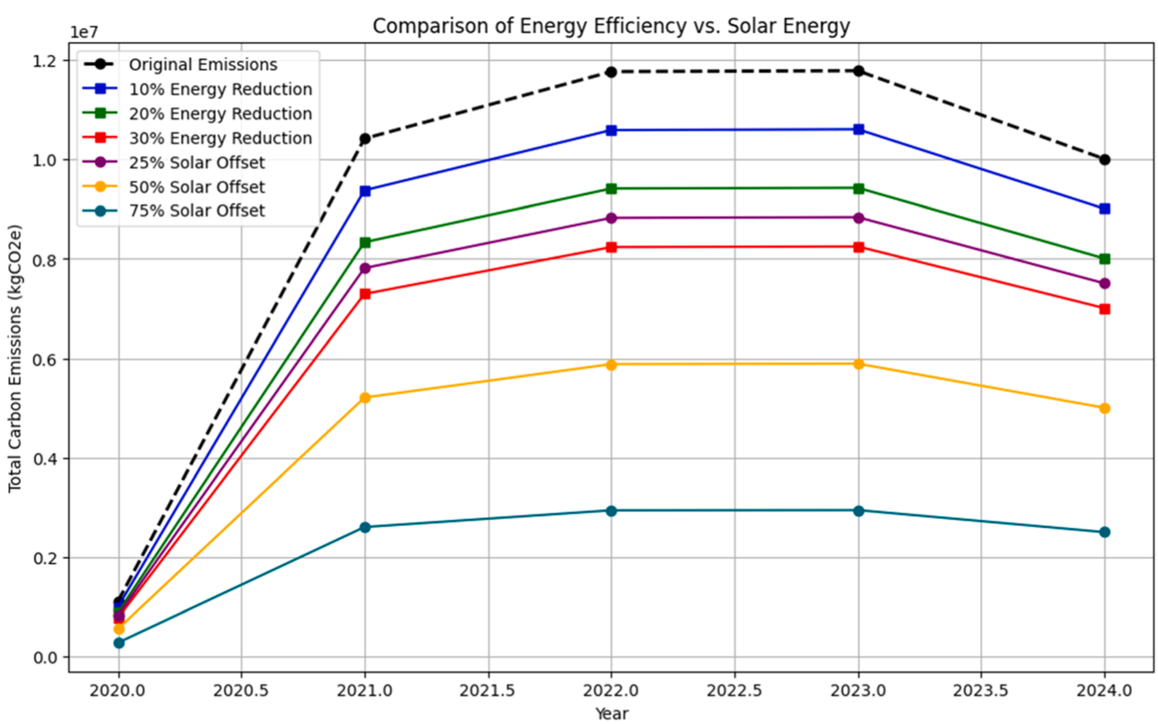

Fig. 7. Comparison of energy efficiency vs. solar energy on carbon emissions.

cooling load empirically confirms that architectural strategies focused on mitigating solar heat gain and enhancing building envelope insulation are paramount. The analysis of building typologies provided deeper insights, revealing significant performance variations. For instance, the high cooling intensity of B-type buildings could be attributed to factors such as less efficient HVAC systems or greater exposure to external heat gains, while the low intensity of the large D-type buildings suggests superior cooling efficiency per unit area. However, the high load per occupant in these same D-type buildings indicates that occupant behavior or internal heat loads are also critical drivers. Furthermore, the complex, non-linear relationship observed between occupant density and cooling intensity highlights its importance as a key predictor variable

Regarding mitigation pathways, the results highlight the complementary role of both demand-side (energy efficiency) and supply-side (solar energy) strategies. While efficiency measures are crucial for reducing overall energy demand, the analysis demonstrates that on-site solar integration is a more powerful lever for deep decarbonization. The finding that a 25 % solar offset surpasses the impact of a 30 % efficiency improvement underscores the importance of prioritizing renewable energy generation.

Beyond technical optimization, future research should explore adaptive behavioral feedback mechanisms that dynamically influence occupant cooling behavior through AI-driven interfaces. Additionally, integrating carbon-aware dispatch strategies—where cooling operations align with real-time grid emissions—could amplify the decarbonization impact. The framework's potential to assess resilience under extreme heat events and its adaptability to low-income housing contexts also warrant further investigation. Finally, coupling SHAP-based model insights with digital twin simulations offers a promising pathway for real-time system calibration and transparent decision-making.

These insights point toward several key strategies for improving the sustainability of district cooling systems. Dynamic control strategies are a crucial first step; smart systems that adjust cooling supply based on real-time data can maintain thermal comfort while minimizing energy waste. These can be complemented by demand-side management (DSM) strategies, such as awareness campaigns and smart thermostats, to encourage occupant conservation. Moreover, implementing AI-powered control systems that adapt to changing conditions can significantly enhance system responsiveness and reduce operational costs. Retrofitting existing buildings with high-efficiency chillers and pumps is another essential step, supported by regular energy audits to identify and track inefficiencies. These strategies, combined with building envelope improvements, provide the tangible pathway to achieving the 30 % energy efficiency scenario modeled in this study. Advanced technologies like digital twins can facilitate real-time simulation and optimization, while the integration of on-site solar PV presents a significant opportunity to offset grid electricity consumption.

Finally, these technical strategies must be supported by the development of comprehensive policies and regulatory frameworks. Financial incentives can accelerate the adoption of energy efficiency retrofits and renewable energy integration. Concurrently, the implementation of stringent, forward-looking building codes that support energy-efficient construction is critical for achieving long-term carbon reduction goals. While this framework provides a clear path forward, its limitations—including the use of monthly data and a specific residential cohort—should be acknowledged. Furthermore, this study lacked detailed data on building envelope materials (e.g., U-values) or specific HVAC system efficiencies. The building stock was constructed around 2008, meaning the analysis reflects the performance of structures from that era; these unobserved factors likely contribute to the remaining variance in cooling load. Future research should aim to incorporate higher-resolution data, expand the analysis to include commercial properties, and conduct a full techno-economic assessment of the proposed interventions.

In conclusion, this research provides a novel, evidence-based framework that moves beyond theoretical models to offer a practical pathway for enhancing the sustainability of district cooling systems. By combining predictive modeling, explainable AI, and comparative scenario analysis, this study demonstrates that a synergistic approach—first optimizing efficiency, then meeting the reduced load with on-site solar energy—offers a scalable blueprint for decarbonizing the urban built environment in the UAE and other arid regions.

CRediT authorship contribution statement

Husna Maliakkal: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation. **Neethu Venugopal:** Writing – review & editing, Supervision, Project administration, Formal analysis. **Kadhim Hayawi:** Supervision, Project administration, Funding acquisition, Conceptualization. **Thanveer Musthafa Hussain:** Validation, Resources, Data curation. **Gomathi Bhavani Rajagopalan:** Validation, Resources, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author agreement statement

We the undersigned declare that manuscript **[EGYR-D-25–03580]** is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We understand that the Corresponding Author is the sole contact for the Editorial process. He/she is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs

Signed by all authors as follows:

Data Availability

The data that has been used is confidential.

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